Data-Driven Investment Intelligence Using NIFTY-50 Market Data

Technical Report

Platform: NIFTY-50 Investment Intelligence Platform — a decision-support system that turns historical OHLCV data into forecasts, portfolios, risk analytics, anomaly flags and explainable BUY/HOLD/SELL recommendations. Dataset: NIFTY-50 Stock Market Data (Kaggle, rohanrao/nifty50-stock-market-data) only. No live data, APIs, news or sentiment. Scope of this report: 49 stocks, daily data spanning 2007-11-27 to 2021-04-30 (~3,300 trading days per stock).

> *Disclaimer: All analytics are derived exclusively from the supplied historical dataset. Forecasts are validated walk-forward; nothing here is investment advice.*

1. Exploratory Data Analysis

Each stock provides daily Open, High, Low, Close, Volume and Turnover, joined to company metadata (name, sector, industry). Ingestion validates the schema, drops non-positive prices and High < Low rows, removes duplicate dates, forward-fills short price gaps (≤5 days) and zero-fills volume gaps, then writes per-symbol Parquet plus a wide close-price matrix.

Key observations from the EDA pass (figures below, generated by ml/eda.py):

- Long-run trends are dominated by a small set of compounders (financials, IT, consumer); price levels span three orders of magnitude across the universe, which motivates scale-free model features (Section 2).
- Return distributions are fat-tailed and leptokurtic — daily returns show far more extreme moves than a Gaussian, validating the use of historical (non-parametric) VaR/CVaR rather than a normal approximation.
- Volatility clusters in time (calm and turbulent regimes alternate), most visibly around the 2008 and 2020 drawdowns — the basis for the anomaly module.
- Sectors diverge materially in risk/return; correlations are high within sectors (banking, IT) and lower across them, which is what makes diversification and risk-parity weighting worthwhile.

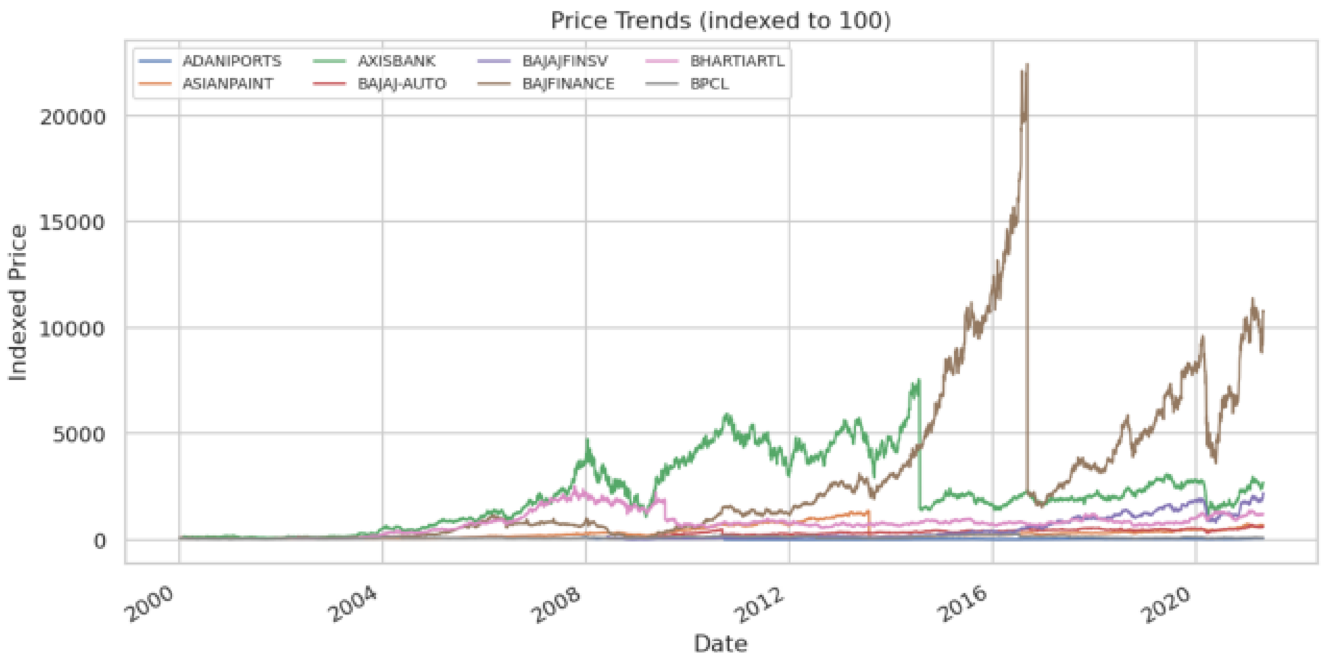

Figure: Price trends across the universe

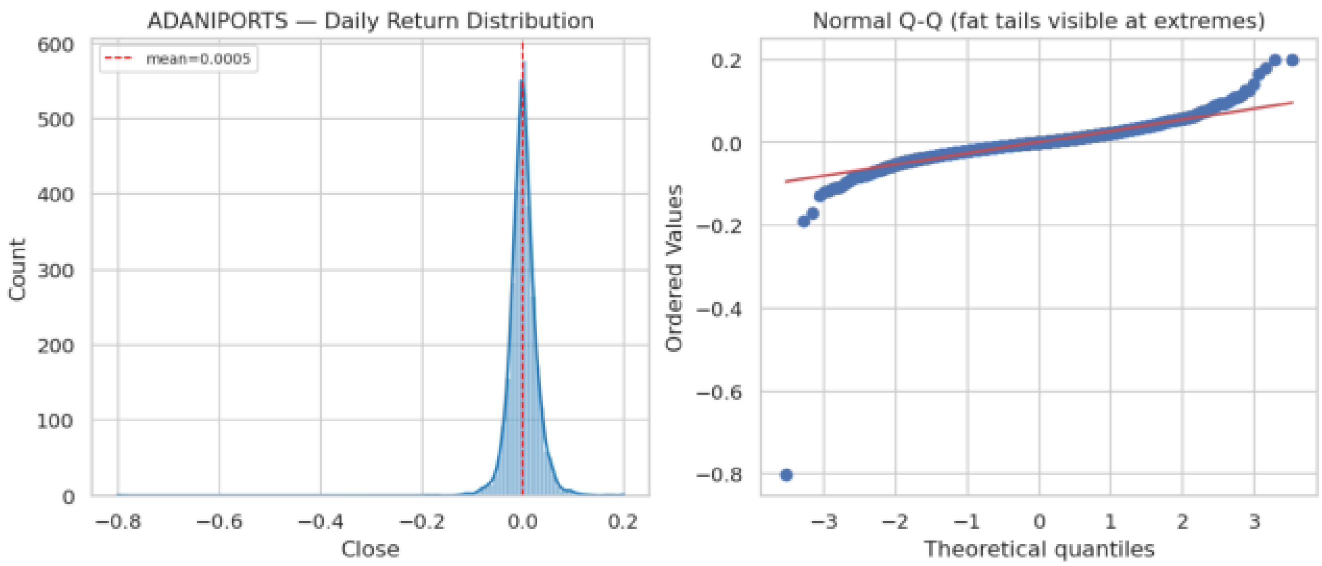

Figure: Daily return distribution (fat tails)

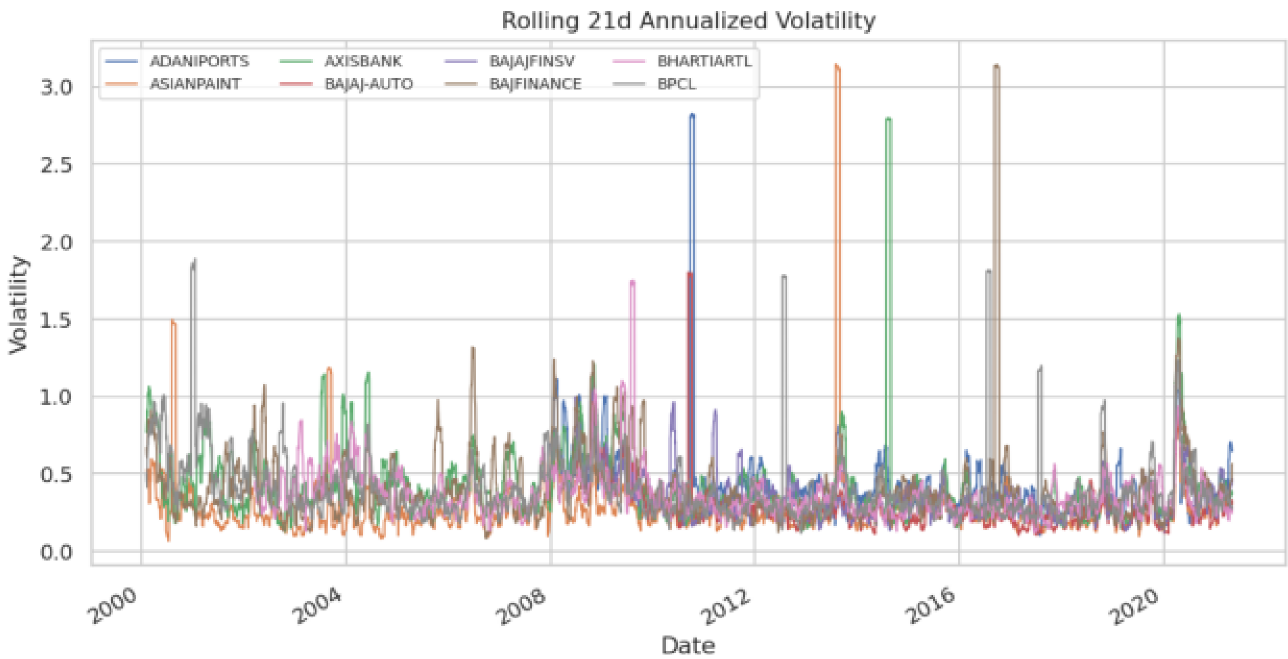

Figure: Rolling volatility and clustering

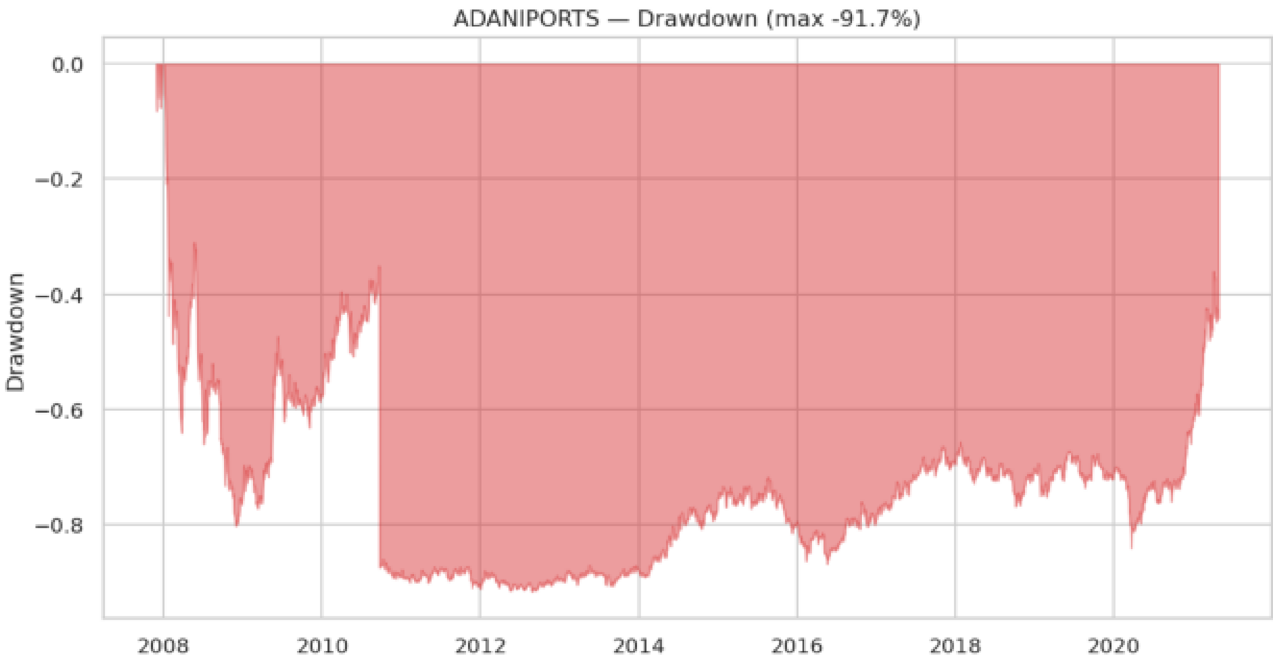

Figure: Drawdown profile

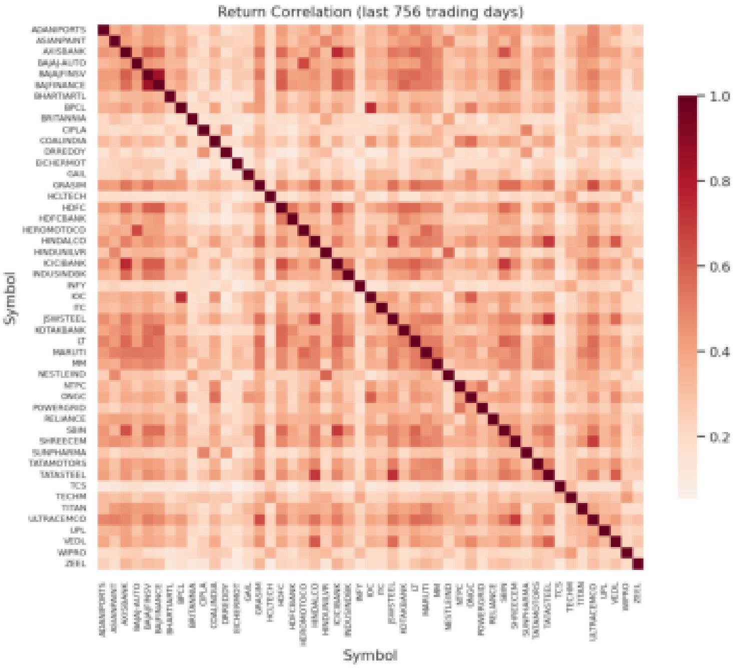

Figure: Cross-stock correlation heatmap

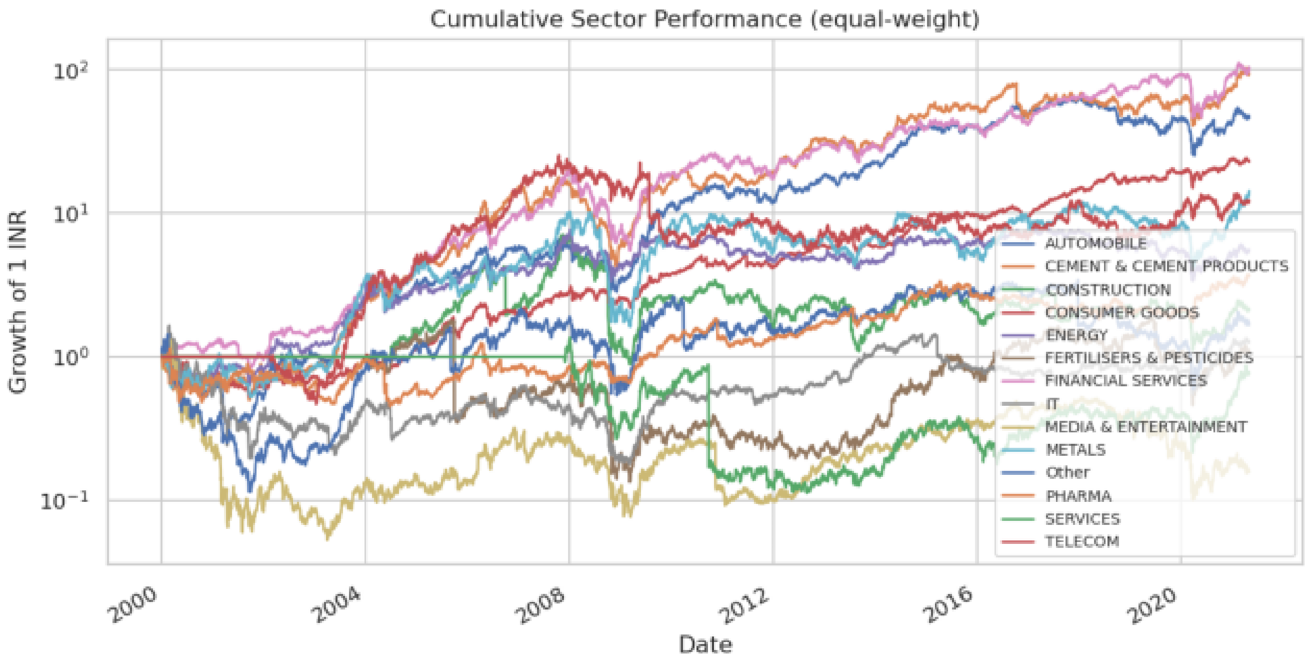

Figure: Sector comparison

2. Feature Engineering

All model inputs are scale-free by construction (ratios and oscillators, never raw price levels) so a single model design transfers across stocks priced from ₹50 to ₹50,000. Features are computed in ml/features.py:

- Trend: close_over_ma{5,10,20,50,200} (price ÷ moving average − 1), MACD line/signal/histogram.
- Momentum: RSI-14, rate-of-change and momentum over 10 and 21 days.

- Volatility: Bollinger %B and bandwidth, ATR% (ATR ÷ close), 21- and 63-day rolling realized volatility.
- Returns: 1/5/21-day simple returns and log returns.
- Volume: volume ratio (vs 20-day average), volume ROC, on-balance-volume direction.
- Relative: sector-relative strength (stock return minus sector-average return).
- Calendar: day-of-week and month encodings to capture seasonality.

Targets. For each stock and horizon $h \in \{1, 5, 20\}$ trading days:

- Regression — forward simple return Close[t+h] / Close[t] − 1.
- Classification — direction (1 if forward return > 0).

3. Methodology

The platform is a set of pure-functional engines over DataFrames, orchestrated by a single memoizing service layer. The end-to-end offline pipeline (scripts/train_all.py) is: ingest → EDA → feature build → train & validate → risk → recommend → report.

Leakage-safe validation is the core methodological commitment (ml/training/):

- Expanding-window time-series CV (default 5 folds) and classic walk-forward splits — the model is only ever trained on the past and tested on the future.
- A gap of h rows is removed at every train/test boundary, because overlapping forward-return labels would otherwise leak future prices into training.
- All reported metrics are out-of-fold only. Regression: RMSE, MAE, MAPE (floored denominator), R^2, directional accuracy. Classification: accuracy, precision, recall, F1, ROC-AUC, directional accuracy.

Reproducibility. Every stochastic component is seeded (random_state=42); optional heavy dependencies (XGBoost, LightGBM, TensorFlow, SHAP, LIME) degrade gracefully so results reproduce on a minimal install. A seeded synthetic generator mirrors the Kaggle schema exactly so the full stack and CI run without the dataset.

4. Model Architecture

Models register only if their library imports (ml/models/registry.py):

Model	Type	Configuration
Linear / Logistic Regression	baseline	standardized pipeline
Random Forest	ensemble	300 trees, depth 8, min-leaf 20 — default API model
XGBoost	boosting	400 trees × depth 5, lr 0.03, subsampled
LightGBM	boosting	500 trees × 31 leaves, lr 0.03

LSTM	deep	64→32 stacked, lookback 30, Huber loss (optional TF)
Transformer	deep	2 encoder blocks, d_model 32, learned positions (optional)

Out-of-fold directional accuracy (fraction of days the sign of the move is predicted correctly) rises with horizon, consistent with daily returns being near-unpredictable while multi-week drift is more learnable:

Horizon	Mean dir-acc (all models)	Best dir-acc
1 day	0.506	0.535
5 days	0.514	0.566
20 days	0.522	0.635

R^2 on raw returns is near zero or negative across models — expected and honestly reported: point-forecasting daily returns is close to a random walk. The platform therefore treats forecasts as one weighted signal among five (Section 7) rather than as standalone price predictions, which is the intended decision-support framing. Deep models (LSTM/Transformer) were benchmarked on a representative subset (HDFCBANK, INFY, RELIANCE, TCS); the LSTM is competitive on 20-day directional accuracy but does not categorically beat gradient-boosted trees, so trees remain the default.

5. Portfolio Construction Logic

Implemented in ml/portfolio/. Inputs are annualized from daily returns; expected returns are shrunk 50% toward the cross-sectional mean because raw historical means are too noisy for mean-variance optimization. The covariance matrix is ridge-regularized for numerical stability. Five long-only optimizers (SLSQP, per-position caps): equal weight, risk parity (fixed-point equal-risk-contribution), Markowitz utility, max Sharpe, min volatility.

Three investor profiles map directly to the problem statement:

Profile	Universe filter	Method	Max weight	Rationale
Conservative	low-volatility 60%	min volatility	12%	capital preservation, tight caps
Balanced	full universe	risk parity	20%	equal risk contribution, diversified
Aggressive	momentum 60%	max Sharpe	30%	return maximization, concentrated

Each recommendation is justified quantitatively: the API returns the allocation plus the portfolio's expected return, volatility and Sharpe. A 5-question questionnaire maps a user to a profile, and ml/backtest.py provides a historical equity-curve backtest with periodic rebalancing for validation.

6. Risk Assessment Methodology

Implemented in ml/risk/. All metrics annualize at 252 trading days with a 6% risk-free rate (Indian 10y G-Sec ballpark, overridable via RISK_FREE_RATE).

- Volatility — annualized standard deviation of daily returns.
- Sharpe / Sortino — excess return over total vs downside deviation.
- Calmar — annual return ÷ max drawdown.
- Maximum drawdown — largest peak-to-trough wealth decline.
- VaR / CVaR — historical (non-parametric), 95%, one-day; CVaR is the mean of the tail beyond VaR.
- Alpha / Beta — CAPM versus an equal-weight universe proxy, since the dataset contains no index series (a stated assumption).

Across the 49-stock universe the mean Sharpe is ≈0.27, with wide dispersion — financials and consumer names carry the highest risk-adjusted returns, cyclicals the deepest drawdowns (e.g. some names exceed −90% peak-to-trough over the full history). These tables feed both the portfolio optimizer and the recommendation engine.

7. Explainability Techniques

Implemented in ml/explainability/. Every model-driven output is attributable:

- SHAP — TreeExplainer for tree ensembles, LinearExplainer for linear models, sampled KernelExplainer otherwise — gives per-feature contributions for any forecast.
- LIME — a second, local-surrogate lens.
- Signed feature importance — a dependency-free fallback (tree impurity or |coefficient|) so explanations are always available.
- Confidence — probability margin for classifiers, or a mapping of out-of-sample directional accuracy for regressors — never in-sample fit.

The recommendation engine is itself transparent: score = 0.30·forecast + 0.20·momentum + 0.20·trend + 0.15·risk_adj + 0.15·sector_rel, each component tanh-squashed to [−1, 1]. BUY ≥ +0.20, SELL ≤ −0.20, else HOLD. Human-readable reasoning is assembled from the strongest components, e.g. "BUY JSWSTEEL because price is trading above its long-term trend, it is outperforming its sector and momentum is strong." A grounded chat assistant answers only from these computed artifacts — no external LLM, fully traceable.

Anomaly detection (ml/anomaly/) adds an unsupervised lens: Isolation Forest (1% contamination), DBSCAN (auto-eps, noise = anomaly) and an autoencoder (Keras, or PCA reconstruction fallback) vote over 7 engineered features; ≥2 votes flags a day, then a rule labels it a volatility spike, unusual return, volume surge or extreme drawdown.

8. Key Insights

- Horizon matters more than model. Directional accuracy climbs from ~0.51 (1-day) to as high as 0.64 (20-day); the choice of horizon dominates the choice of algorithm. Daily price prediction is effectively a random walk and is treated as such.
- Returns are not point-predictable — but decisions are supportable. Negative R^2 on returns is reported honestly; value comes from combining a modest forecast edge with momentum, trend, risk-adjustment and sector-relative signals into an auditable score, not from a single magic prediction.
- Risk is highly dispersed. Mean Sharpe ≈0.27 hides a wide spread; profile-based construction (min-vol / risk-parity / max-Sharpe) lets the same universe serve very different investors with quantitatively justified allocations.
- Current snapshot. On the latest data the full five-signal engine (forecast included) flags 23 BUY, 20 HOLD, 6 SELL across the universe — a constructive but not indiscriminate stance, with every call backed by its component breakdown. (The dashboard's fast default view omits the forecast signal for responsiveness; ?with_forecasts=true reproduces the figures here.)
- Everything is explainable and reproducible. Seeded, leakage-safe, dependency-graceful, and runnable end-to-end via scripts/train_all.py or one-command Docker — directly targeting the explainability, transparency and reproducibility criteria.

Assumptions & limitations. Equal-weight universe proxy stands in for a market index; only ~13 years of the available history are used per stock after cleaning; no transaction-cost or liquidity modeling beyond the backtest; forecasts carry genuine uncertainty and are decision-support inputs, not advice.